NIHR | National Institute for Health and Care Research

The post-award effort of managing and reporting on funded research: a scoping review

K Crane, A Blatch-Jones, K Fackrell

NIHR Coordinating Centre, School of Healthcare Enterprise and Innovation, University of Southampton, UK.

Research on Research

BACKGROUND

The availability of research funds is declining and subject to tighter compliance, performance, and fiscal controls. As funders are under more pressure to monitor the research they fund, the effort of **post-award management** is also increasing.

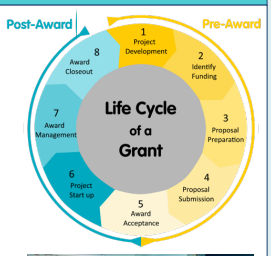
Post-award, researchers and Higher Education Institutions (HEIs) are required **to report to funders** on the progress, outputs, outcomes, and impacts of research. However, inconsistent reporting requirements and a growing demand for research information can lead to unnecessary administrative effort that impacts on academic culture, decision-making, and research activity.

Identifying this **unnecessary effort,** and the **areas in post-award management needing improvement**, is crucial if funders and HEIs are to streamline their practices and provide better support with reporting activities.

This scoping review aimed to answer the following questions:

1. What does the evidence say about processes and effort in post-award management, the roles of funders and HEIs, and the purpose of reporting on research?

2. Where is effort perceived as unnecessary and how can funders and HEIs improve processes for all stakeholders?



METHODS

Scoping reviews help identify research priorities, clarify concepts, and provide background information for further research. A systematic approach is used to scope the wide literature base (and identify evidence gaps) to answer: *"what does the literature say?"* and *"what has been done previously?"*

Reporting frameworks: PRISMA-ScR and JBI

Post-award management:
Definition: processes related to the awarding, set up, management, monitoring, and reporting of externally funded research
Context: All funded academic research (UK and international).
Participants: Funders, HEIs, researchers, administrators, evaluators (and wider)
Inclusion criteria: Descriptive evidence of processes, tasks or considerations involved in post-award management (any academic research field)
Exclusion criteria: Lacking above or out of study scope (e.g., fiscal management, auditing)
Record types: Peer-reviewed publications, grey literature (e.g., blogs), reports, conference abstracts, presentations, and websites.
Search strategy: No limit on publication date, status, or country of origin. Database searches were combined with manual searches of journals and funder websites.
Databases: Medline, Embase, PubMed, Web of Science, Google Scholar (March 2022-23)
Funder websites: 11 websites (March 2022-23):

Alzheimer's Research | CIHR IRSC | erc European Research Council | hrc nz | UKRI Medical Research Council | NIH National Institutes of Health | NHMRC | National Research Foundation | UGC | wellcome | NIHR National Institute for Health and Care Research

RESULTS

Evidence was extracted from **52 records** and **11 websites (Fig 1).**

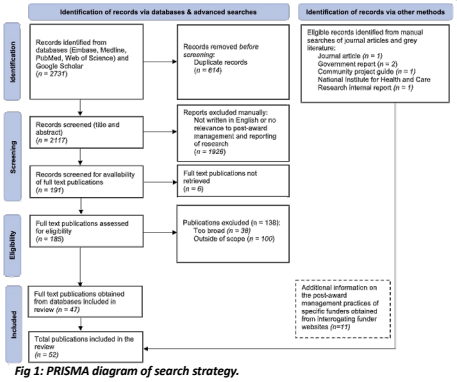


Fig 1: PRISMA diagram of search strategy.

Characteristics: 51% research/reviews and 86% UK/US-focused.
Extracted data: Funder processes (50%), HEI processes (19%), both (30%); stakeholders (85%), digital systems (54%); terminology.

What does the literature say about post-award management? (n=52)

The **process** involves...

Component (14 total)	*Examples* of processes/tasks ('items')
Award set-up, compliance and close-out (27 items)	Contract negotiations, hiring, IRB, compliance Grant inactivation, invention statement
Financial management and reporting (18 items)	Accounting, monitoring award balances Payroll, staff expenses
Progress reporting (22 items)	Milestones and deliverables Preliminary results, achievements, unexpected events
End-of-grant reporting (15 items)	Plain English summary Outputs, impact statement
Digital tracking of outputs, outcomes and impacts (8 items)	Outputs, impact statement PPI Question set (Researchfish)
Variation requests (10 items)	Changes to research, plan, PI, HEI, commencement date Extension requests
In-person and informal reporting (5 items)	Site visits and reviews Researcher surveys (e.g., for impact assessment)
Performance reporting (5 items)	Participant enrolment/recruitment Research income, higher degrees awarded
Data management and accessibility (5 items)	Data management plan, data teams Completion of institutional data repositories
Promoting, publishing, and disseminating research	Selecting journals (OA compliance) Social media, digital platform, community dissemination etc
Clinical trial transparency (3 items)	Trial registration Reporting registry number, outputs/outcomes to funder
Security of research (2 items)	Reporting foreign influence, COI Data security
Feedback to funders (1 item)	Written feedback for programme improvement
Assessing researcher performance (1 item)	Researcher CV (achievements and affiliations)

For HEIs and researchers, most of the **post-award effort** is in the set up and reporting of research awards.

The **purpose** of the process is...

Theme (14 total)	*Examples* of purpose-related statements (57 total)
Research impact assessment	• To demonstrate translation and societal impact • To connect research with outputs and impacts (e.g. policy)
Compliance	• To comply with funder and HEI policies/legislation • To ensure proper use of funds for research
Accountability	• To account for public spending and demonstrate payback • To build trust between funders, HEIs and the public
Programme development	• For funders to understand successes and failures • For funders to identify gaps where funding is needed
Responsible research conduct	• To maintain integrity and ethics in research • To protect the welfare of research participants
Transparency	• To report non-academic publications (e.g., policy briefings) • To help inform the work of other organisations
Securing future funds	• To advocate for future funding support • To build a relationship with the funder
Monitoring projects	• Monitor progress against milestones and course-correct • For researchers to reflect on their progress/discoveries
Protecting reputations	• To raise the profiles of researchers, projects and HEIs • To prevent consequences of non-compliance
Supporting researchers	• To capture mentoring activities • To understand how to better support researchers
Reusing information	• To reduce duplication and waste of research • To share data and learning between funders and HEIs
Protecting research from theft	• To assess and mitigate potential security risks • To prevent COI or breach of research regulations
Promoting innovation	• To indicate that innovative research has been conducted
Improving research management	• To determine how research processes can be improved for researchers and HEIs

Areas of Post-Award Management needing improvement (n=29):

Area (6 total)	Examples of issues (e.g., processes, unnecessary effort, research culture)	Solutions being applied or recommendations
Supporting researchers	1. Ineffective working relationships between researchers and research administrators (RAs) impact on research success 2. Compliance and grant management (e.g., progress reports) take up a significant amount of funded research time (~40%)	1. Facilitate better RA-researcher working relationships through communications, collaboration and mutual trust 2. Grant-related processes should be streamlined and administrative support provided to free up valuable research time
Research impact assessment (RIA)	1. There is inconsistency and a lack of transparency in how some funders and countries approach RIA 2.Increased reliance on digital impact tracking creates duplication of effort (too many systems) and leads to research waste (less time for research)	1. Funders should adopt effective RIA frameworks and be more transparent (e.g., about how the reported information affects researchers) 2. Information should be better shared and re-used and digital systems more interoperable to reduce burden on researchers
Data management and accessibility	1. HEIs store vast quantities of data but it is hard to curate and find externally, limiting its re-use value 2. HEIs often receive multiple requests for the same information (e.g., compliance, performance) due to siloed approaches to data management	1. Data curation centres (e.g., RDRDS) can help HEIs better manage their data and track research outputs 2. Standardisation of data (e.g., universal key standards, identifiers) can reduce manual effort in information sharing and reporting

Funder practices (n=11)

Funders' approaches to post-award management currently vary significantly, most notably in:

- The terminology used to refer to post-award processes,
- The reporting requirements and the frequency of reporting
- The type of information requested and where it must be reported
- The digital systems used for award management and reporting.
- The level of guidance for researchers and HEIs
- Information on policies (and why information is asked for, how it is used, and who uses it).

CONCLUSIONS

- The need for post-award management and reporting is widely appreciated. However, the effort can be considerable.
- More rigorous evidence and stakeholder consultations are needed so that barriers to efficient research delivery can be identified and addressed more collaboratively in a connected research environment.
- HEIs and researchers could benefit from more administrative support services and guidance on 'impact' and training in post-award management.
- Reducing duplication and research waste in reporting is key to minimising effort whilst increasing value to research end-users.

RECOMMENDATIONS

For funders: (examples)

- Involve the sector in deciding monitoring and reporting approaches
- Clarify what information is asked for, why it is needed, and who uses it.
- Collect feedback from researchers, including 'metrics' of effort or burden
- Streamline compliance and assurance requirements to reduce duplication and delay to the research

For HEIs: (examples)

- Provide more research management and administrative support
- Share information with other HEIs and standardise processes
- Collaborate with funders and suppliers to improve interoperability of systems for better data sharing and re-use
- Provide regular feedback to funders on how to improve processes

University of Southampton

K.Crane@soton.ac.uk
@KsenCrane
Dr Ksenia Crane
0000-0002-8471-2165

The work is supported by the National Institute for Health and Care Research (NIHR) as part of the NIHR Coordinating Centre. The views expressed are those of the authors and not necessarily those of the NIHR or the Department of Health and Social Care.

Publication: *Crane, Blatch-Jones and Fackrell. The post-award effort of managing and reporting on funded research: a scoping review. F1000Research 2023, 12:863*

Images/ illustrations: wellcome.org, chemistryworld.org, genproresearch.org, and kyinbridges.com